

1. Questions

Study the following information carefully and answer the questions given below:

There are 4 turtles in point P, and they are going somewhere in the sea, and they return back to point P. The speed of turtle A is 75% of that of turtle B, and the total time taken by turtle A to go downstream and return back to point P is 24 hours. The distance travelled by turtle C in downstream is 4 km more than that travelled by turtle B in upstream. The ratio of the downstream speed to the upstream speed of turtle D is 3:1, respectively, and the ratio of the speeds of turtles C and D is 7:10, respectively. The time taken by turtle C to travel upstream is 5 times the time taken by turtle D to travel downstream.

Notes:

- i). The speed of the stream is 5 km/hr.
- ii). All turtles started from point P in a downstream direction to different directions and distances.
- iii). The distance travelled by turtle A in downstream is 4 km less than that by turtle B in upstream.
- iv). The speed of turtle A is 60% of that of turtle D.

Turtle A travels 50% more distance than the distance it travels downstream. The total time taken by turtle A to go and return to point P is 'n' hours. Which of the following statements is true?

- I:** The value of 'n' is a perfect square.
- II:** The value of 'n' is a perfect cube
- III:** The value of 'n' is a prime number.

- a. Only I
- b. Only II
- c. Only III
- d. Both I and II
- e. Both I and III

2. Questions

The total time taken by turtles A and B to travel downstream together is what percentage of the total time taken by turtles C and D to travel upstream together?

- a. 11.11%
- b. 9.09%
- c. 16.66%
- d. 33.33%
- e. 27.27%

3. Questions

If the speed of turtle C is increased by 14.28% and speed of the stream is increased by 'm%', then the time taken by turtle C to travel the distance (mentioned in the direction) downstream is 30 minutes earlier. Find the value of 'm'.

- a. 20
- b. 40
- c. 25
- d. 10
- e. 50

4. Questions

The speed of turtle E is the average speed of turtles B and D together. The turtle E travels downstream in 5 hours less than that of upstream. The sum of time taken by the turtles D and E to travel downstream together is ____ hours.

- a. 4 hours
- b. 5 hours 30 minutes
- c. 6 hours 15 minutes
- d. 4 hours 30 minutes
- e. 5 hours

5. Questions

Which of the following statements is true?

K = Total time taken by turtle C to cover the downstream distance and half of the upstream distance.

L = Total time taken by turtle B to cover the downstream distance and the upstream distance.

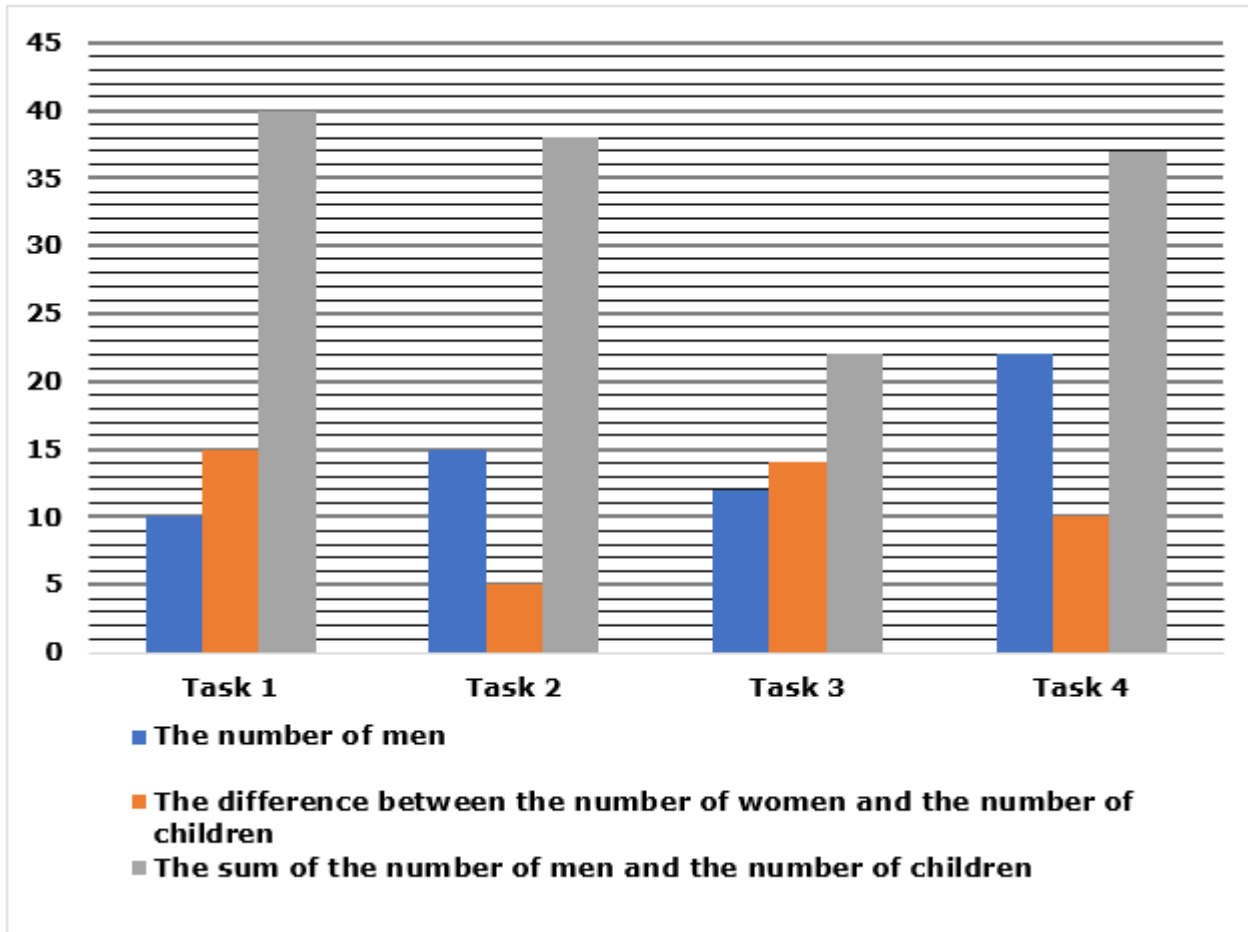
- a. $L > K$
- b. $K \geq L$
- c. $k = L$
- d. $L < K$
- e. $L \geq K$

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6. Questions

Study the following information carefully and answer the questions given below:

The bar graph below shows the number of men, the difference between the number of women and children, and the sum of the number of men and children in four different tasks.



Notes:

- i). The number of men and women in tasks 1 and 2 are consecutive multiples of 5 and consecutive multiples of 3, respectively.
- ii). The number of women in task 3 is twice the number of men in that task.
- iii). The efficiency ratio of one man, one woman, and one child in task 1 is 7 : 6 : 2, and in task 2 it is 9 : 7 : 3.
- iv). The total work completed in tasks 1 and 2 by men, women, and children together in one day is 220 units and 330 units, respectively. The total work completed in task 4 in one day (by men, women, and children together) is 2.5 times the work completed in task 1.
- v). One woman completes twice the number of units in task 4 as compared to task 1, and one child completes the same number of units in tasks 1 and 4.

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If one man completes 10 units per day in task 4, find which of the following statements is true according to the number of women in task 4.

I: Perfect square

II: Multiple of 5

III: Odd number.

- a. Only I
- b. Both (I) and (II)

- c. Both (II) and (III)
- d. Only II
- e. All of them

7. Questions

In task 2, both men and children start the work together. After 'k' (day or days), both men and children leave the work, and the remaining work is completed by the women in 1 day. Find the value of 'k'. (The total work of this task is the same as mentioned in the direction)

- a. 1
- b. 2
- c. 3
- d. 5
- e. 4

8. Questions

If the units completed per day by a woman in task 3 are 66.66% of the units completed per day by a man in task 3, and the units completed per day by a man in task 3 are 66.66% more than those in task 2, then the total work completed in task 3 by men, women, and children together in one day is 80 units less than that in task 4. What is the efficiency ratio of men, women, and children in task 3?

- a. 3: 4: 2
- b. 3: 2: 1
- c. 4: 5: 1
- d. 5: 6: 3
- e. 4: 6: 2

9. Questions

If in task 1, 20% of the men, 40% of the women, and 30% of the children left, then the units of work in task 1 completed by the remaining men, women, and children together in one day is approximately what percentage of the total units of work in task 1 completed by all men, women, and children together in one day?

- a. 63%
- b. 59%
- c. 73%
- d. 69%
- e. 79%

10. Questions

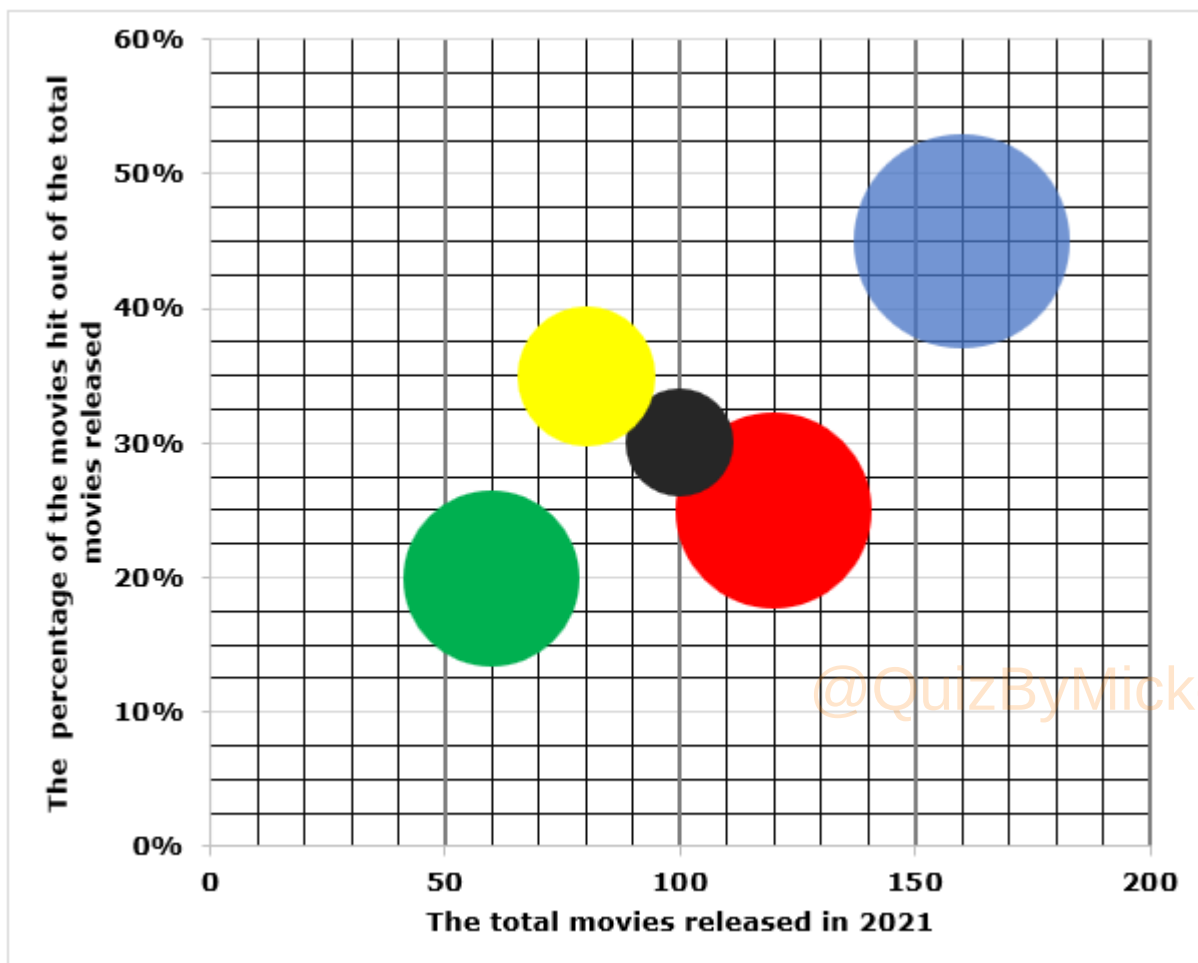
If the number of men and women in task 5 are in the ratio 7:9, and one man and one woman complete 4 units and 3 units per day in task 5, respectively, and the total units of work in tasks 1 and 5 are the same, then the time taken by men and women together to complete task 5 is 2 days. What is the number of men working on task 5?

- a. 12
- b. 18
- c. 14
- d. 10
- e. 21

11. Questions

Study the following information carefully and answer the questions given below:

The bubble graph below shows the total movies released in 2021 and the percentage of the hit movies out of the total movies released in 2021 in five different industries.



Notes:

i). The red, black, yellow, blue, and green colors represent the five film industries A, B, C, D, and E, respectively.

ii). The total movies released in 2021 = The number of hit movies released in 2021 + The number of flop movies released in 2021

iii). The number of movies released in 2022 in five different industries is 25% less, 28% more, 15% less, 22.5% more, and 5% more than the previous year, respectively.

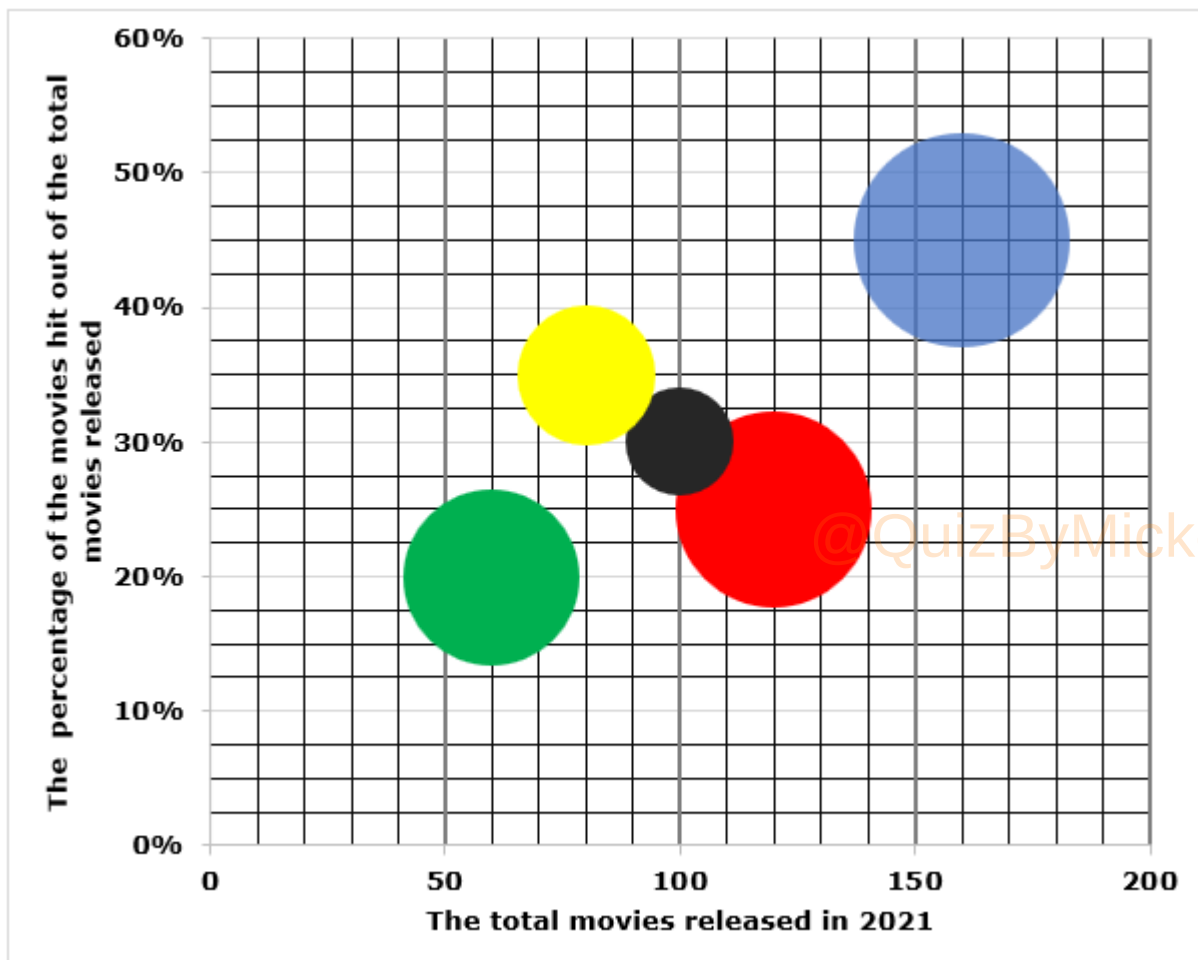
In 2021, in film industry A, the number of comedy movies released is 20%. The number of horror movies released is 33.33% of the remaining movies, the number of action movies released is 75% of the next remaining movies, and the rest are adventure movies. Find the difference between the number of adventure movies released in 2021 and the number of flop movies released in 2021.

- a. 74
- b. 68
- c. 64
- d. 72
- e. 56

12. Questions

Study the following information carefully and answer the questions given below:

The bubble graph below shows the total movies released in 2021 and the percentage of the movies hit out of the total movies released in 2021 in five different industries.



Notes:

- i). The red, black, yellow, blue, and green colors represent the five film industries A, B, C, D, and E, respectively.
- ii). The total movies released in 2021 = The number of hit movies released in 2021 + The number of flop movies released in 2021
- iii). The number of movies released in 2022 in five different industries is 25% less, 28% more, 15% less, 22.5% more, and 5% more than the previous year, respectively.

If the percentages of hit movies released in 2022 in industries B and E are 25% and 22.22% of the total movies released in 2022 in their respective industries, then the total number of flop movies released in 2022 in industries B and E together is what percentage of the total movies released in industries A and C together in 2021?

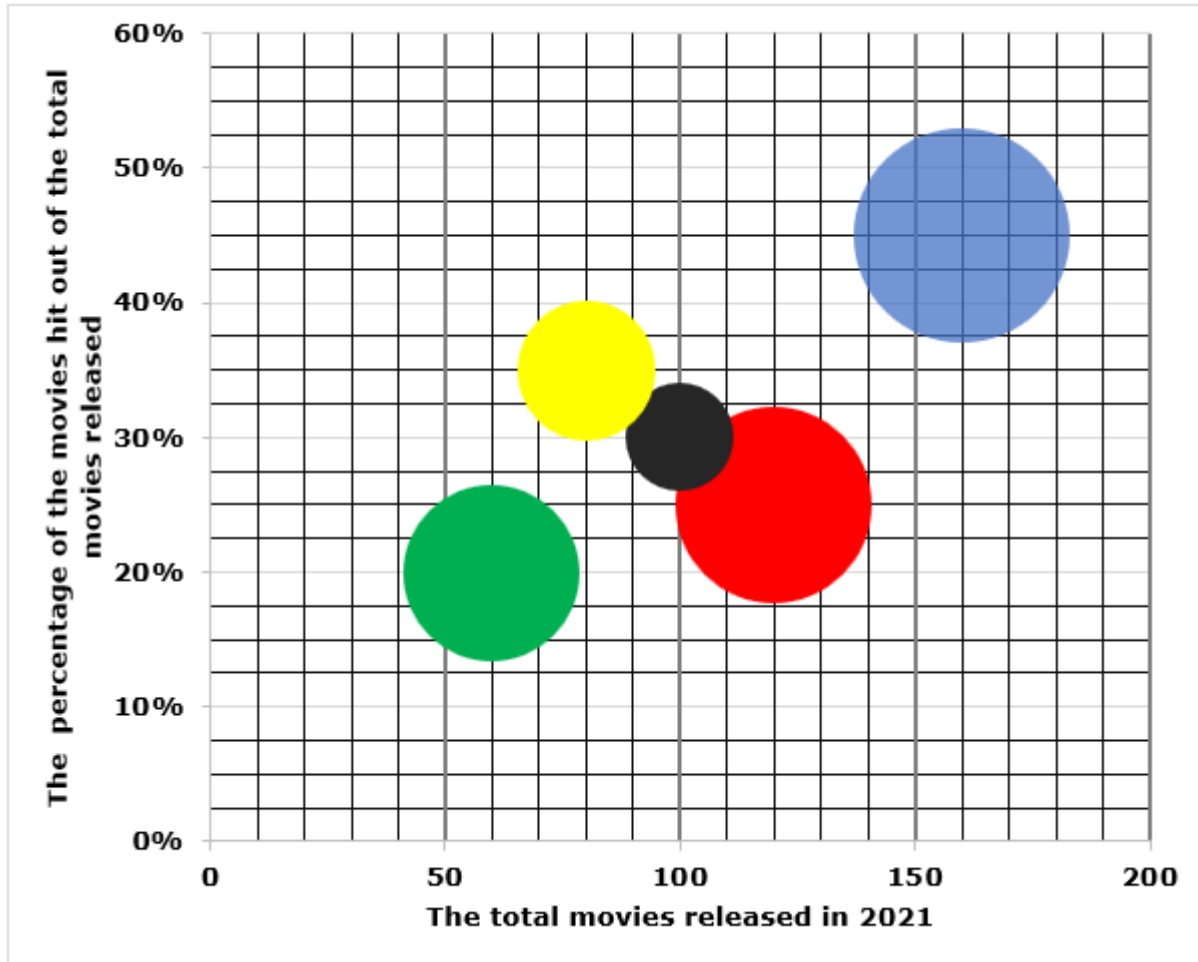
- a. 67.5%
- b. 77.5%
- c. 72.5%
- d. 62.5%
- e. 82.5%

13. Questions

Study the following information carefully and answer the questions given below:

The bubble graph below shows the total movies released in 2021 and the percentage of the hit movies out of the total movies released in 2021 in five different industries.

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Notes:

- The red, black, yellow, blue, and green colors represent the five film industries A, B, C, D, and E, respectively.
- The total movies released in 2021 = The number of hit movies released in 2021 + The number of flop movies released in 2021
- The number of movies released in 2022 in five different industries is 25% less, 28% more, 15% less, 22.5% more, and 5% more than the previous year, respectively.

What is the ratio of the total number of hit movies released in 2021 in all industries to the total number of flop movies released in 2021 in all industries?

- 21: 32
- 43: 83
- 23: 33
- 43: 87
- 41: 81

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14. Questions

If the ratio of hit movies to flop movies released in industry C in 2022 is 10:7, and the number of flop movies released in industry C is 40% of that in industry D in 2022, what is the average number

of hit movies released in industry D in 2021 and 2022 together?

- a. 93
- b. 99
- c. 87
- d. 88
- e. 84

15. Questions

If the number of hit movies released in industry F in 2021 is 75% more than that in industry E, and the total movies released in industry F in 2021 is the average of the total movies released in industries C, D, and E in 2021, find the number of flop movies released in industry F in 2021.

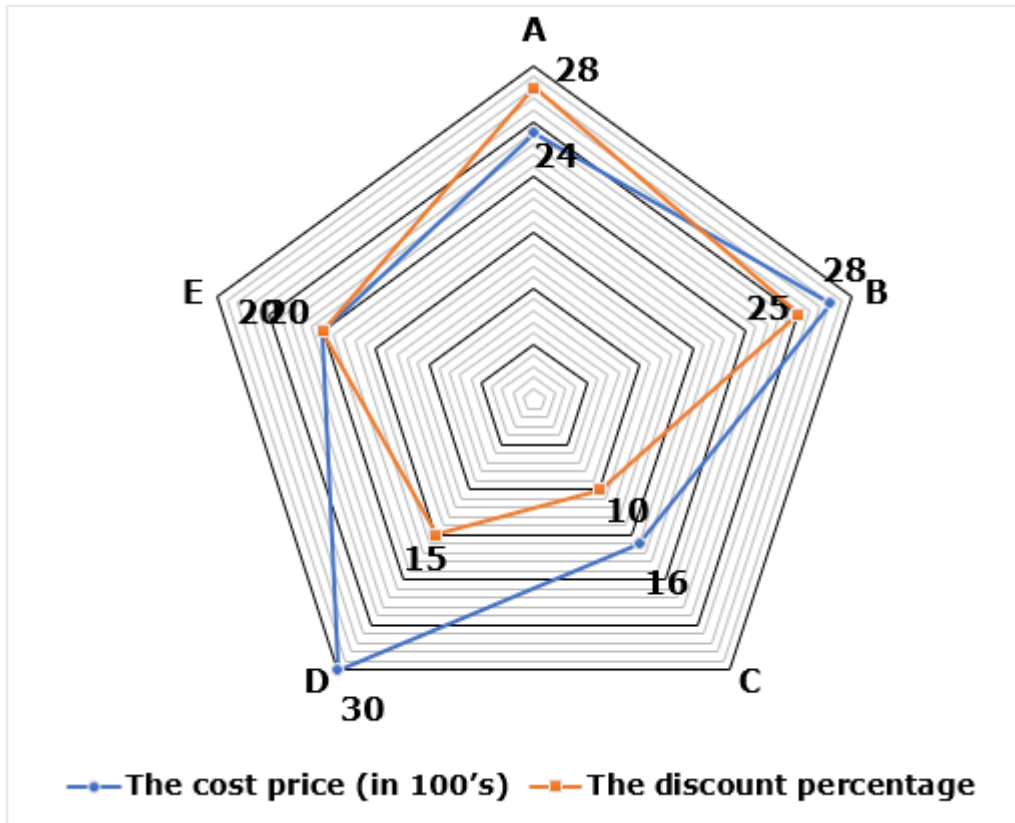
- a. 69
- b. 71
- c. 73
- d. 63
- e. 79

16. Questions

Study the following information carefully and answer the questions given below:

The radar graph below shows the cost price (in hundred's) and the discount percentage of the five different items.

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Note: Cost price(in hundreds)

The table below shows the profit/loss percentage of the five different items. (The positive and negative signs indicate the profit and loss percentage respectively.)

Items	The profit/loss percentage
A	-X%
B	+Y%
C	-(X/10)%
D	+(Y + 33)%
E	+(Y + 12)%

Note:

i). The profit amount of item B is Rs. 560.

If the ratio of values of X and Y is 1:2, respectively. Find the difference between the selling price of items A and B.

- Rs. 1500
- Rs. 1200
- Rs. 1800
- Rs. 900
- Rs. 1000

17. Questions

What is the ratio of the marked-up percentage of item D to that of item E?

- a. 15: 11
- b. 17: 13
- c. 13: 11
- d. 16: 11
- e. 16: 13

18. Questions

If the discount amount of item C is 27.5% of that of item E. Find the marked-up percentage of item A.

- a. 25%
- b. 15%
- c. 20%
- d. 40%
- e. 50%

19. Questions

If the cost price of item F is Y% more than that of item C and the marked-up percentage of items D and F are same. If the item F sells at 5% profit. Find the sum of the discount amount of items D and F.

- a. Rs. 2200
- b. Rs. 2100
- c. Rs. 2250
- d. Rs. 2150
- e. Rs. 2350

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20. Questions

The sum of the marked price of the all five items together is Rs. 6140 more than that of cost price. The marked price of item A is Rs. 1240 more than that of item C. Find the loss amount of items A and C together.

- a. Rs. 256
- b. Rs. 222
- c. Rs. 196

d. Rs. 284

e. Rs. 272

Explanations:

1. Questions

The speed of the stream = 5 km/hr

The ratio of the downstream speed to the upstream speed of turtle D is 3:1, respectively

Let us assume that the speed of turtle D is 'x' km/hr

The downstream speed of turtle D = $(x + 5)$ km/hr

The upstream speed of turtle D = $(x - 5)$ km/hr

Let

$$(x + 5)/(x - 5) = 3/1$$

$$x + 5 = 3x - 15$$

$$2x = 20$$

$$x = 10$$

The downstream speed of turtle D = $10 + 5 = 15$ km/hr

The upstream speed of turtle D = $10 - 5 = 5$ km/hr

The ratio of the speeds of turtles C and D is 7:10, respectively.

The speed of turtle C = $(10/10) * 7 = 7$ km/hr

The downstream speed of turtle C = $7 + 5 = 12$ km/hr

The upstream speed of turtle C = $7 - 5 = 2$ km/hr

The time taken by turtle C to travel upstream is 5 times the time taken by turtle D to travel downstream.

Let assume the distance travelled by turtle D in downstream is 'd' km and the distance travelled by turtle C in upstream is 'c' km.

$$c/2 = 5 * (d/15)$$

$$c/d = 2/3$$

$$c : d = 2 : 3 = 2y : 3y$$

The distance travelled by turtle C in downstream or upstream = $2y$ km

The distance travelled by turtle D in downstream or upstream = $3y$ km

Let

The distance travelled by turtle C in downstream is 4 km more than that travelled by turtle B in upstream.

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The distance travelled by turtle B in upstream = $(2y - 4)$ km

The distance travelled by turtle A in downstream is 4 km less than that travelled by turtle B in upstream.

The distance travelled by turtle A in downstream or upstream = $2y - 4 - 4 = (2y - 8)$ km

The speed of turtle A is 60% of that of turtle D.

The speed of turtle A = 60% of 10 = 6 km/hr

The downstream speed of turtle A = $6 + 5 = 11$ km/hr

The upstream speed of turtle A = $6 - 5 = 1$ km/hr

Then

The speed of turtle A is 75% of that of turtle B

The speed of turtle B = $(6 * 100/75) = 8$ km/hr

The downstream speed of turtle B = $8 + 5 = 13$ km/hr

The upstream speed of turtle B = $8 - 5 = 3$ km/hr

Let

The total time taken by turtle A to go downstream and return back to point P be 24 hours.

$$(2y - 8)/11 + (2y - 8)/1 = 24$$

$$(2y - 8 + 22y - 88)/11 = 24$$

$$(24y - 96)/11 = 24$$

$$(y - 4)/11 = 1$$

$$y - 4 = 11$$

$$y = 15$$

Let

The distance travelled by turtle C in downstream or upstream = $2y = 30$ km

The distance travelled by turtle D in downstream or upstream = $3y = 35$ km

The distance travelled by turtle B in downstream or upstream = $30 - 4 = 26$ km

The distance travelled by turtle A in downstream or upstream = $30 - 8 = 22$ km

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Turtles	The speed of turtle (in km/hr)	The downstream speed (in km/hr)	The upstream speed (in km/hr)	The distance travelled (downstream/upstream) in km
A	6	11	1	22
B	8	13	3	26
C	7	12	2	30
D	10	15	5	45

Answer: A

Let

According to the question

The turtle A travels a new distance in downstream = 150% of 22 = 33 km

The total time taken by turtle A to return to point P = $n = (33/11) + (33/1) = 3 + 33 = 36$ hours

$n = 36$

The value of 'n' is a perfect square.

2. Questions

The speed of the stream = 5 km/hr

The ratio of the downstream speed to the upstream speed of turtle D is 3:1, respectively

Let us assume that the speed of turtle D is 'x' km/hr

The downstream speed of turtle D = $(x + 5)$ km/hr

The upstream speed of turtle D = $(x - 5)$ km/hr

Let

$$(x + 5)/(x - 5) = 3/1$$

$$x + 5 = 3x - 15$$

$$2x = 20$$

$$x = 10$$

The downstream speed of turtle D = $10 + 5 = 15$ km/hr

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The ratio of the speeds of turtles C and D is 7:10, respectively.

The speed of turtle C = $(10/10) * 7 = 7$ km/hr

The downstream speed of turtle C = $7 + 5 = 12$ km/hr

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The time taken by turtle C to travel upstream is 5 times the time taken by turtle D to travel downstream.

Let assume the distance travelled by turtle D in downstream is 'd' km and the distance travelled by turtle C in upstream is 'c' km.

$$c/2 = 5 * (d/15)$$

$$c/d = 2/3$$

$$c : d = 2 : 3 = 2y : 3y$$

The distance travelled by turtle C in downstream or upstream = $2y$ km

The distance travelled by turtle D in downstream or upstream = $3y$ km

Let

The distance travelled by turtle C in downstream is 4 km more than that travelled by turtle B in upstream.

The distance travelled by turtle B in upstream = $(2y - 4)$ km

The distance travelled by turtle A in downstream is 4 km less than that travelled by turtle B in upstream.

The distance travelled by turtle A in downstream or upstream = $2y - 4 - 4 = (2y - 8)$ km

The speed of turtle A is 60% of that of turtle D.

The speed of turtle A = 60% of 10 = 6 km/hr

The downstream speed of turtle A = $6 + 5 = 11$ km/hr

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Then

The speed of turtle A is 75% of that of turtle B

The speed of turtle B = $(6 * 100/75) = 8$ km/hr

The downstream speed of turtle B = $8 + 5 = 13$ km/hr

The upstream speed of turtle B = $8 - 5 = 3$ km/hr

Let

The total time taken by turtle A to go downstream and return back to point P be 24 hours.

$$(2y - 8)/11 + (2y - 8)/1 = 24$$

$$(2y - 8 + 22y - 88)/11 = 24$$

$$(24y - 96)/11 = 24$$

$$(y - 4)/11 = 1$$

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$$y = 15$$

Let

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The distance travelled by turtle C in downstream or upstream = $2y = 30$ km

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The distance travelled by turtle A in downstream or upstream = $30 - 8 = 22$ km

Turtles	The speed of turtle (in km/hr)	The downstream speed (in km/hr)	The upstream speed (in km/hr)	The distance travelled (downstream/upstream) in km
A	6	11	1	22
B	8	13	3	26
C	7	12	2	30
D	10	15	5	45

Answer: C

The total time taken by turtles A and B to travel downstream together = $(22/11) + (26/13) = 2 + 2 = 4$ hours

The total time taken by turtles C and D to travel upstream together = $(30/2) + (45/5) = 15 + 9 = 24$ hours

Thus, percentage = $(4/24) * 100 = 16.66\%$

3. Questions

The speed of the stream = 5 km/hr

The ratio of the downstream speed to the upstream speed of turtle D is 3:1, respectively

Let us assume that the speed of turtle D is 'x' km/hr

The downstream speed of turtle D = $(x + 5)$ km/hr

The upstream speed of turtle D = $(x - 5)$ km/hr

Let

$$(x + 5)/(x - 5) = 3/1$$

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$$2x = 20$$

$$x = 10$$

The downstream speed of turtle D = $10 + 5 = 15$ km/hr

The upstream speed of turtle D = $10 - 5 = 5$ km/hr

The ratio of the speeds of turtles C and D is 7:10, respectively.

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The downstream speed of turtle C = $7 + 5 = 12$ km/hr

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The upstream speed of turtle C = $7 - 5 = 2$ km/hr

The time taken by turtle C to travel upstream is 5 times the time taken by turtle D to travel downstream.

Let assume the distance travelled by turtle D in downstream is 'd' km and the distance travelled by turtle C in upstream is 'c' km.

$$c/2 = 5 * (d/15)$$

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The distance travelled by turtle A in downstream or upstream = $2y - 4 - 4 = (2y - 8)$ km

The speed of turtle A is 60% of that of turtle D.

The speed of turtle A = 60% of 10 = 6 km/hr

The downstream speed of turtle A = $6 + 5 = 11$ km/hr

The upstream speed of turtle A = $6 - 5 = 1$ km/hr

Then

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The downstream speed of turtle B = $8 + 5 = 13$ km/hr

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Let

The total time taken by turtle A to go downstream and return back to point P be 24 hours.

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@QuizByMickey

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Turtles	The speed of turtle (in km/hr)	The downstream speed (in km/hr)	The upstream speed (in km/hr)	The distance travelled (downstream/upstream) in km
A	6	11	1	22
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C	7	12	2	30
D	10	15	5	45

Answer: B

The new speed of turtle C = 114.28% of 7 = $(8/7) * 7 = 8$ km/hr

Let us assume that the new downstream speed = 'z' km/hr

Let

Distance = 30 km

$(30/12) - (30/z) = 30/60$

$2.5 - (30/z) = 0.5$

$30/z = 2$

$z = 15$

The new downstream speed = 15 km/hr

The new speed of stream = $15 - 8 = 7$ km/hr

So,

$m\% = [(7 - 5)/5] * 100 = 40\%$

$m = 40$

4. Questions

The speed of the stream = 5 km/hr

The ratio of the downstream speed to the upstream speed of turtle D is 3:1, respectively

Let us assume that the speed of turtle D is 'x' km/hr

The downstream speed of turtle D = $(x + 5)$ km/hr

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The upstream speed of turtle D = $(x - 5)$ km/hr

Let

$$(x + 5)/(x - 5) = 3/1$$

$$x + 5 = 3x - 15$$

$$2x = 20$$

$$x = 10$$

The downstream speed of turtle D = $10 + 5 = 15$ km/hr

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The distance travelled by turtle A in downstream or upstream = $2y - 4 - 4 = (2y - 8)$ km

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The speed of turtle A = 60% of 10 = 6 km/hr

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Then

The speed of turtle A is 75% of that of turtle B

The speed of turtle B = $(6 * 100/75) = 8$ km/hr

The downstream speed of turtle B = $8 + 5 = 13$ km/hr

The upstream speed of turtle B = $8 - 5 = 3$ km/hr

Let

The total time taken by turtle A to go downstream and return back to point P be 24 hours.

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$$(2y - 8 + 22y - 88)/11 = 24$$

$$(24y - 96)/11 = 24$$

$$(y - 4)/11 = 1$$

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$$y = 15$$

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The distance travelled by turtle C in downstream or upstream = $2y = 30$ km

The distance travelled by turtle D in downstream or upstream = $3y = 35$ km

The distance travelled by turtle B in downstream or upstream = $30 - 4 = 26$ km

The distance travelled by turtle A in downstream or upstream = $30 - 8 = 22$ km

Turtles	The speed of turtle (in km/hr)	The downstream speed (in km/hr)	The upstream speed (in km/hr)	The distance travelled (downstream/upstream) in km
A	6	11	1	22
B	8	13	3	26
C	7	12	2	30
D	10	15	5	45

Answer: E

The speed of turtle E = $(8 + 10)/2 = 9$ km/hr

The speed of the stream = 5 km/hr

Let us assume that the distance = 's' km

$$[s/(9 - 5)] - [s/(9 + 5)] = 5$$

$$(s/4) - (s/14) = 5$$

$$(14s - 4s)/56 = 5$$

$$10s/56 = 5$$

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$$s = 28$$

The time taken by turtle E downstream = $28/14 = 2$ hours

The time taken by turtle D downstream = $45/15 = 3$ hours

Thus, sum = $2 + 3 = 5$ hours

5. Questions

The speed of the stream = 5 km/hr

The ratio of the downstream speed to the upstream speed of turtle D is 3:1, respectively

Let us assume that the speed of turtle D is 'x' km/hr

The downstream speed of turtle D = $(x + 5)$ km/hr

The upstream speed of turtle D = $(x - 5)$ km/hr

Let

$$(x + 5)/(x - 5) = 3/1$$

$$x + 5 = 3x - 15$$

$$2x = 20$$

$$x = 10$$

The downstream speed of turtle D = $10 + 5 = 15$ km/hr

The upstream speed of turtle D = $10 - 5 = 5$ km/hr

The ratio of the speeds of turtles C and D is 7:10, respectively.

The speed of turtle C = $(10/10) * 7 = 7$ km/hr

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The upstream speed of turtle C = $7 - 5 = 2$ km/hr

The time taken by turtle C to travel upstream is 5 times the time taken by turtle D to travel downstream.

Let assume the distance travelled by turtle D in downstream is 'd' km and the distance travelled by turtle C in upstream is 'c' km.

$$c/2 = 5 * (d/15)$$

$$c/d = 2/3$$

$$c : d = 2 : 3 = 2y : 3y$$

The distance travelled by turtle C in downstream or upstream = $2y$ km

The distance travelled by turtle D in downstream or upstream = $3y$ km

Let

The distance travelled by turtle C in downstream is 4 km more than that travelled by turtle B in upstream.

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The distance travelled by turtle B in upstream = $(2y - 4)$ km

The distance travelled by turtle A in downstream is 4 km less than that travelled by turtle B in upstream.

The distance travelled by turtle A in downstream or upstream = $2y - 4 - 4 = (2y - 8)$ km

The speed of turtle A is 60% of that of turtle D.

The speed of turtle A = 60% of 10 = 6 km/hr

The downstream speed of turtle A = $6 + 5 = 11$ km/hr

The upstream speed of turtle A = $6 - 5 = 1$ km/hr

Then

The speed of turtle A is 75% of that of turtle B

The speed of turtle B = $(6 \times 100/75) = 8$ km/hr

The downstream speed of turtle B = $8 + 5 = 13$ km/hr

The upstream speed of turtle B = $8 - 5 = 3$ km/hr

Let

The total time taken by turtle A to go downstream and return back to point P be 24 hours.

$$(2y - 8)/11 + (2y - 8)/1 = 24$$

$$(2y - 8 + 22y - 88)/11 = 24$$

$$(24y - 96)/11 = 24$$

$$(y - 4)/11 = 1$$

$$y - 4 = 11$$

$$y = 15$$

Let

The distance travelled by turtle C in downstream or upstream = $2y = 30$ km

The distance travelled by turtle D in downstream or upstream = $3y = 35$ km

The distance travelled by turtle B in downstream or upstream = $30 - 4 = 26$ km

The distance travelled by turtle A in downstream or upstream = $30 - 8 = 22$ km

@QuizByMickey

Turtles	The speed of turtle (in km/hr)	The downstream speed (in km/hr)	The upstream speed (in km/hr)	The distance travelled (downstream/upstream) in km
A	6	11	1	22
B	8	13	3	26
C	7	12	2	30
D	10	15	5	45

Answer: A

$$K = (30/12) + (15/2)$$

$$K = 2.5 + 7.5$$

$$K = 10$$

And

$$L = (26/13) + (26/3)$$

$$L = 2 + (8 \frac{2}{3})$$

$$L = (10 \frac{2}{3}) \text{ or } 10.67$$

Thus, $L > K$

6. Questions

Task 1

The number of men = 10

The sum of the number of men and the number of children = 40

The number of children = $40 - 10 = 30$

The difference between the number of women and the number of children = 15

If (women > children)

The number of women - the number of children = 15

The number of women - 30 = 15

The number of women = 45

The number of men and women in task 1 is a consecutive multiple of 5

10 and 45 are not consecutive multiples of 5

If (women < children)

The number of children - the number of women = 15

30 - The number of women = 15

@QuizByMickey

The number of women = 15

The number of men and women in task 1 is a consecutive multiple of 5

10 and 15 are consecutive multiples of 5

So, the number of women = 15

Let

The efficiency ratio of the man, woman, and child in task 1 is 7: 6: 2 ($7x$: $6x$: $2x$) and the total units of task 1 completed by men, women, and children together in 1 day is 220

$$[(10 * 7x) + (15 * 6x) + (30 * 2x)] * 1 = 220$$

$$70x + 90x + 60x = 220$$

$$220x = 220$$

$$x = 1$$

$$\text{A man completed units per day} = 7 * 1 = 7$$

$$\text{A woman completed units per day} = 6 * 1 = 6$$

$$\text{A child completed units per day} = 2 * 1 = 2$$

Task 2

The number of men = 15

The sum of the number of men and the number of children = 38

$$\text{The number of children} = 38 - 15 = 23$$

The difference between the number of women and the number of children = 5

If (women > children)

$$\text{The number of women} - \text{the number of children} = 5$$

$$\text{The number of women} - 23 = 5$$

$$\text{The number of women} = 28$$

The number of men and women in task 2 is a consecutive multiple of 3

15 and 28 are not consecutive multiples of 3

If (women < children)

$$\text{The number of children} - \text{the number of women} = 5$$

$$23 - \text{The number of women} = 5$$

$$\text{The number of women} = 18$$

The number of men and women in task 2 is a consecutive multiple of 3

15 and 18 are consecutive multiples of 3

@QuizByMickey

So, The number of women = 18

Let

The efficiency ratio of the man, woman, and child in task 2 is 9: 7: 3 (9y: 7y: 3y) and the total units of task 2 completed by men, women, and children together in 1 day is 330

$$[(15 * 9y) + (18 * 7y) + (23 * 3y)] * 1 = 330$$

$$135y + 126y + 69y = 330$$

$$330y = 330$$

$$y = 1$$

$$\text{A man completed units per day} = 9 * 1 = 9$$

$$\text{A woman completed units per day} = 7 * 1 = 7$$

$$\text{A child completed units per day} = 3 * 1 = 3$$

Task 3

$$\text{The number of men} = 12$$

$$\text{The number of women} = 12 * 2 = 24$$

$$\text{The sum of the number of men and the number of children} = 22$$

$$\text{The number of children} = 22 - 12 = 10$$

The efficiency ratio of man, woman and child is not known.

Task 4

$$\text{The number of men} = 22$$

$$\text{The sum of the number of men and the number of children} = 37$$

$$\text{The number of children} = 37 - 22 = 15$$

$$\text{The difference between the number of women and the number of children} = 10$$

If (women > children)

$$\text{The number of women} - \text{the number of children} = 10$$

$$\text{The number of women} - 15 = 10$$

$$\text{The number of women} = 25$$

If (women < children)

$$\text{The number of children} - \text{the number of women} = 10$$

$$15 - \text{The number of women} = 10$$

$$\text{The number of women} = 5$$

The units of work completed by one woman in task 4 is two times that of task 1, and the units of work completed by one child in task 1 and task 4 are equal.

@QuizByMickey

A woman completed units per day = $6 * 2 = 12$

A child completed units per day = 2

In one day, the total units of Task 4 completed by men, women and children together was 2.5 times that of Task 1.

In one day, the total units of Task 4 completed by men, women and children = $2.5 * 220 = 550$ units.

Let table

Task	Man		Woman		Child		Total work completed in 1 day (units)
	Total	Efficiency (units)	Total	Efficiency (units)	Total	Efficiency (units)	
1	10	7	15	6	30	2	220
2	15	9	18	7	23	3	330
3	12	-	24	-	10	-	-
4	22	-	25/5	12	15	2	550

Answer: E

According to the question,

Units completed by man per day in task 4 = 10

WKT, The number of women = 5 or 25

If (5)

Total work completed in one day = $(22 * 10) + (5 * 12) + (15 * 2)$

= $220 + 60 + 30$

= 310

$310 \neq 550$

If (25)

Total work completed in one day = $(22 * 10) + (25 * 12) + (15 * 2)$

= $220 + 300 + 30$

= 550

So, the number of women in task 4 = 25

7. Questions

Task 1

The number of men = 10

The sum of the number of men and the number of children = 40

The number of children = $40 - 10 = 30$

@QuizByMickey

The difference between the number of women and the number of children = 15

If (women > children)

The number of women - the number of children = 15

The number of women - 30 = 15

The number of women = 45

The number of men and women in task 1 is a consecutive multiple of 5

10 and 45 are not consecutive multiples of 5

If (women < children)

The number of children - the number of women = 15

30 - The number of women = 15

The number of women = 15

The number of men and women in task 1 is a consecutive multiple of 5

10 and 15 are consecutive multiples of 5

So, the number of women = 15

Let

The efficiency ratio of the man, woman, and child in task 1 is 7: 6: 2 (7x: 6x: 2x) and the total units of task 1 completed by men, women, and children together in 1 day is 220

$$[(10 * 7x) + (15 * 6x) + (30 * 2x)] * 1 = 220$$

$$70x + 90x + 60x = 220$$

$$220x = 220$$

$$x = 1$$

$$\text{A man completed units per day} = 7 * 1 = 7$$

$$\text{A woman completed units per day} = 6 * 1 = 6$$

$$\text{A child completed units per day} = 2 * 1 = 2$$

Task 2

@QuizByMickey

The number of men = 15

The sum of the number of men and the number of children = 38

The number of children = 38 - 15 = 23

The difference between the number of women and the number of children = 5

If (women > children)

The number of women - the number of children = 5

The number of women – 23 = 5

The number of women = 28

The number of men and women in task 2 is a consecutive multiple of 3

15 and 28 are not consecutive multiples of 3

If (women < children)

The number of children - the number of women = 5

23 - The number of women = 5

The number of women = 18

The number of men and women in task 2 is a consecutive multiple of 3

15 and 18 are consecutive multiples of 3

So, The number of women = 18

Let

The efficiency ratio of the man, woman, and child in task 2 is 9: 7: 3 (9y: 7y: 3y) and the total units of task 2 completed by men, women, and children together in 1 day is 330

$$[(15 * 9y) + (18 * 7y) + (23 * 3y)] * 1 = 330$$

$$135y + 126y + 69y = 330$$

$$330y = 330$$

$$y = 1$$

$$\text{A man completed units per day} = 9 * 1 = 9$$

$$\text{A woman completed units per day} = 7 * 1 = 7$$

$$\text{A child completed units per day} = 3 * 1 = 3$$

Task 3

The number of men = 12

The number of women = 12 * 2 = 24

The sum of the number of men and the number of children = 22 @QuizByMickey

The number of children = 22 – 12 = 10

The efficiency ratio of man, woman and child is not known.

Task 4

The number of men = 22

The sum of the number of men and the number of children = 37

The number of children = 37 – 22 = 15

The difference between the number of women and the number of children = 10

If (women > children)

The number of women - the number of children = 10

The number of women – 15 = 10

The number of women = 25

If (women < children)

The number of children - the number of women = 10

15 - The number of women = 10

The number of women = 5

The units of work completed by one woman in task 4 is two times that of task 1, and the units of work completed by one child in task 1 and task 4 are equal.

A woman completed units per day = $6 * 2 = 12$

A child completed units per day = 2

In one day, the total units of Task 4 completed by men, women and children together was 2.5 times that of Task 1.

In one day, the total units of Task 4 completed by men, women and children = $2.5 * 220 = 550$ units.

Let table

Task	Man		Woman		Child		Total work completed in 1 day (units)
	Total	Efficiency (units)	Total	Efficiency (units)	Total	Efficiency (units)	
1	10	7	15	6	30	2	220
2	15	9	18	7	23	3	330
3	12	-	24	-	10	-	-
4	22	-	25/5	12	15	2	550

Answer: A

@QuizByMickey

Total work = 330 units

$$[15 * 9 + 23 * 3] * k + (18 * 7) * 1 = 330$$

$$204k + 126 = 330$$

$$204k = 204$$

$$k = 1$$

8. Questions

Task 1

The number of men = 10

The sum of the number of men and the number of children = 40

The number of children = $40 - 10 = 30$

The difference between the number of women and the number of children = 15

If (women > children)

The number of women - the number of children = 15

The number of women - 30 = 15

The number of women = 45

The number of men and women in task 1 is a consecutive multiple of 5

10 and 45 are not consecutive multiples of 5

If (women < children)

The number of children - the number of women = 15

30 - The number of women = 15

The number of women = 15

The number of men and women in task 1 is a consecutive multiple of 5

10 and 15 are consecutive multiples of 5

So, the number of women = 15

Let

The efficiency ratio of the man, woman, and child in task 1 is 7: 6: 2 (7x: 6x: 2x) and the total units of task 1 completed by men, women, and children together in 1 day is 220

$$[(10 * 7x) + (15 * 6x) + (30 * 2x)] * 1 = 220$$

$$70x + 90x + 60x = 220$$

$$220x = 220$$

$$x = 1$$

A man completed units per day = $7 * 1 = 7$

A woman completed units per day = $6 * 1 = 6$

A child completed units per day = $2 * 1 = 2$

Task 2

The number of men = 15

The sum of the number of men and the number of children = 38

The number of children = $38 - 15 = 23$

@QuizByMickey

The difference between the number of women and the number of children = 5

If (women > children)

The number of women - the number of children = 5

The number of women - 23 = 5

The number of women = 28

The number of men and women in task 2 is a consecutive multiple of 3

15 and 28 are not consecutive multiples of 3

If (women < children)

The number of children - the number of women = 5

23 - The number of women = 5

The number of women = 18

The number of men and women in task 2 is a consecutive multiple of 3

15 and 18 are consecutive multiples of 3

So, The number of women = 18

Let

The efficiency ratio of the man, woman, and child in task 2 is 9: 7: 3 (9y: 7y: 3y) and the total units of task 2 completed by men, women, and children together in 1 day is 330

$$[(15 * 9y) + (18 * 7y) + (23 * 3y)] * 1 = 330$$

$$135y + 126y + 69y = 330$$

$$330y = 330$$

$$y = 1$$

$$\text{A man completed units per day} = 9 * 1 = 9$$

$$\text{A woman completed units per day} = 7 * 1 = 7$$

$$\text{A child completed units per day} = 3 * 1 = 3$$

Task 3

The number of men = 12

The number of women = $12 * 2 = 24$

The sum of the number of men and the number of children = 22

The number of children = $22 - 12 = 10$

The efficiency ratio of man, woman and child is not known.

Task 4

@QuizByMickey

The number of men = 22

The sum of the number of men and the number of children = 37

The number of children = $37 - 22 = 15$

The difference between the number of women and the number of children = 10

If (women > children)

The number of women - the number of children = 10

The number of women - 15 = 10

The number of women = 25

If (women < children)

The number of children - the number of women = 10

15 - The number of women = 10

The number of women = 5

The units of work completed by one woman in task 4 is two times that of task 1, and the units of work completed by one child in task 1 and task 4 are equal.

A woman completed units per day = $6 * 2 = 12$

A child completed units per day = 2

In one day, the total units of Task 4 completed by men, women and children together was 2.5 times that of Task 1.

In one day, the total units of Task 4 completed by men, women and children = $2.5 * 220 = 550$ units.

Let table

Task	Man		Woman		Child		Total work completed in 1 day (units)
	Total	Efficiency (units)	Total	Efficiency (units)	Total	Efficiency (units)	
1	10	7	15	6	30	2	220
2	15	9	18	7	23	3	330
3	12	-	24	-	10	-	-
4	22	-	25/5	12	15	2	550

Answer: B

Units completed per day by a man in task 3 = 166.66% of 9 = $(5/3) * 9 = 15$

Units completed per day by a woman in task 3 = 66.66% of 15 = $(2/3) * 15 = 10$

Units completed per day by children in task 3 = p

Total task 3 completed by men, women, and children in one day is = $550 - 80 = 470$

Let

$$[12 * 15 + 24 * 10 + 10 * p] * 1 = 470$$

$$180 + 240 + 10p = 470$$

$$10p = 50$$

$$p = 5$$

Thus, the efficiency ratio of the men, women, and children in task 3

$$= 15: 10: 5$$

$$= 3: 2: 1$$

9. Questions

Task 1

The number of men = 10

The sum of the number of men and the number of children = 40

The number of children = $40 - 10 = 30$

The difference between the number of women and the number of children = 15

If (women > children)

The number of women - the number of children = 15

The number of women - 30 = 15

The number of women = 45

The number of men and women in task 1 is a consecutive multiple of 5

10 and 45 are not consecutive multiples of 5

If (women < children)

The number of children - the number of women = 15

30 - The number of women = 15

The number of women = 15

The number of men and women in task 1 is a consecutive multiple of 5

10 and 15 are consecutive multiples of 5

So, the number of women = 15

Let

The efficiency ratio of the man, woman, and child in task 1 is 7: 6: 2 ($7x: 6x: 2x$) and the total units of task 1 completed by men, women, and children together in 1 day is 220

$$[(10 * 7x) + (15 * 6x) + (30 * 2x)] * 1 = 220$$

$$70x + 90x + 60x = 220$$

$$220x = 220$$

$$x = 1$$

$$\text{A man completed units per day} = 7 * 1 = 7$$

$$\text{A woman completed units per day} = 6 * 1 = 6$$

$$\text{A child completed units per day} = 2 * 1 = 2$$

Task 2

$$\text{The number of men} = 15$$

$$\text{The sum of the number of men and the number of children} = 38$$

$$\text{The number of children} = 38 - 15 = 23$$

$$\text{The difference between the number of women and the number of children} = 5$$

If (women > children)

$$\text{The number of women} - \text{the number of children} = 5$$

$$\text{The number of women} - 23 = 5$$

$$\text{The number of women} = 28$$

The number of men and women in task 2 is a consecutive multiple of 3

15 and 28 are not consecutive multiples of 3

If (women < children)

$$\text{The number of children} - \text{the number of women} = 5$$

$$23 - \text{The number of women} = 5$$

$$\text{The number of women} = 18$$

The number of men and women in task 2 is a consecutive multiple of 3

15 and 18 are consecutive multiples of 3

So, The number of women = 18

Let

@QuizByMickey

The efficiency ratio of the man, woman, and child in task 2 is 9: 7: 3 (9y: 7y: 3y) and the total units of task 2 completed by men, women, and children together in 1 day is 330

$$[(15 * 9y) + (18 * 7y) + (23 * 3y)] * 1 = 330$$

$$135y + 126y + 69y = 330$$

$$330y = 330$$

$$y = 1$$

A man completed units per day = $9 * 1 = 9$

A woman completed units per day = $7 * 1 = 7$

A child completed units per day = $3 * 1 = 3$

Task 3

The number of men = 12

The number of women = $12 * 2 = 24$

The sum of the number of men and the number of children = 22

The number of children = $22 - 12 = 10$

The efficiency ratio of man, woman and child is not known.

Task 4

The number of men = 22

The sum of the number of men and the number of children = 37

The number of children = $37 - 22 = 15$

The difference between the number of women and the number of children = 10

If (women > children)

The number of women - the number of children = 10

The number of women - 15 = 10

The number of women = 25

If (women < children)

The number of children - the number of women = 10

15 - The number of women = 10

The number of women = 5

The units of work completed by one woman in task 4 is two times that of task 1, and the units of work completed by one child in task 1 and task 4 are equal.

A woman completed units per day = $6 * 2 = 12$

A child completed units per day = 2

In one day, the total units of Task 4 completed by men, women and children together was 2.5 times that of Task 1.

In one day, the total units of Task 4 completed by men, women and children = $2.5 * 220 = 550$ units.

Let table

@QuizByMickey

Task	Man		Woman		Child		Total work completed in 1 day (units)
	Total	Efficiency (units)	Total	Efficiency (units)	Total	Efficiency (units)	
1	10	7	15	6	30	2	220
2	15	9	18	7	23	3	330
3	12	-	24	-	10	-	-
4	22	-	25/5	12	15	2	550

Answer: D

In task 1

Remaining number of men = 80% of 10 = 8

Remaining number of women = 60% of 15 = 9

Remaining number of children = 70% of 30 = 21

The units of work completed by the remaining men, women, and children together in 1 day in task 1

$$= [8 * 7 + 9 * 6 + 21 * 2] * 1$$

$$= 56 + 54 + 42$$

$$= 152$$

$$\text{Thus, percentage} = (152/220) * 100 \approx 69\%$$

10. Questions

Task 1

The number of men = 10

The sum of the number of men and the number of children = 40

The number of children = 40 – 10 = 30

The difference between the number of women and the number of children = 15

If (women > children)

The number of women - the number of children = 15

The number of women – 30 = 15

The number of women = 45

The number of men and women in task 1 is a consecutive multiple of 5

10 and 45 are not consecutive multiples of 5

If (women < children)

The number of children - the number of women = 15

@QuizByMickey

30 - The number of women = 15

The number of women = 15

The number of men and women in task 1 is a consecutive multiple of 5

10 and 15 are consecutive multiples of 5

So, the number of women = 15

Let

The efficiency ratio of the man, woman, and child in task 1 is 7: 6: 2 ($7x$: $6x$: $2x$) and the total units of task 1 completed by men, women, and children together in 1 day is 220

$$[(10 * 7x) + (15 * 6x) + (30 * 2x)] * 1 = 220$$

$$70x + 90x + 60x = 220$$

$$220x = 220$$

$$x = 1$$

$$\text{A man completed units per day} = 7 * 1 = 7$$

$$\text{A woman completed units per day} = 6 * 1 = 6$$

$$\text{A child completed units per day} = 2 * 1 = 2$$

Task 2

The number of men = 15

The sum of the number of men and the number of children = 38

$$\text{The number of children} = 38 - 15 = 23$$

The difference between the number of women and the number of children = 5

If (women > children)

$$\text{The number of women} - \text{the number of children} = 5$$

$$\text{The number of women} - 23 = 5$$

$$\text{The number of women} = 28$$

The number of men and women in task 2 is a consecutive multiple of 3 @QuizByMickey

15 and 28 are not consecutive multiples of 3

If (women < children)

$$\text{The number of children} - \text{the number of women} = 5$$

$$23 - \text{The number of women} = 5$$

$$\text{The number of women} = 18$$

The number of men and women in task 2 is a consecutive multiple of 3

15 and 18 are consecutive multiples of 3

So, The number of women = 18

Let

The efficiency ratio of the man, woman, and child in task 2 is 9: 7: 3 (9y: 7y: 3y) and the total units of task 2 completed by men, women, and children together in 1 day is 330

$$[(15 * 9y) + (18 * 7y) + (23 * 3y)] * 1 = 330$$

$$135y + 126y + 69y = 330$$

$$330y = 330$$

$$y = 1$$

$$\text{A man completed units per day} = 9 * 1 = 9$$

$$\text{A woman completed units per day} = 7 * 1 = 7$$

$$\text{A child completed units per day} = 3 * 1 = 3$$

Task 3

$$\text{The number of men} = 12$$

$$\text{The number of women} = 12 * 2 = 24$$

$$\text{The sum of the number of men and the number of children} = 22$$

$$\text{The number of children} = 22 - 12 = 10$$

The efficiency ratio of man, woman and child is not known.

Task 4

$$\text{The number of men} = 22$$

$$\text{The sum of the number of men and the number of children} = 37$$

$$\text{The number of children} = 37 - 22 = 15$$

$$\text{The difference between the number of women and the number of children} = 10$$

If (women > children)

$$\text{The number of women} - \text{the number of children} = 10$$

$$\text{The number of women} - 15 = 10$$

$$\text{The number of women} = 25$$

If (women < children)

$$\text{The number of children} - \text{the number of women} = 10$$

$$15 - \text{The number of women} = 10$$

$$\text{The number of women} = 5$$

@QuizByMickey

The units of work completed by one woman in task 4 is two times that of task 1, and the units of work completed by one child in task 1 and task 4 are equal.

A woman completed units per day = $6 * 2 = 12$

A child completed units per day = 2

In one day, the total units of Task 4 completed by men, women and children together was 2.5 times that of Task 1.

In one day, the total units of Task 4 completed by men, women and children = $2.5 * 220 = 550$ units.

Let table

Task	Man		Woman		Child		Total work completed in 1 day (units)
	Total	Efficiency (units)	Total	Efficiency (units)	Total	Efficiency (units)	
1	10	7	15	6	30	2	220
2	15	9	18	7	23	3	330
3	12	-	24	-	10	-	-
4	22	-	25/5	12	15	2	550

Answer: C

The ratio of the number of men and women in task 5 = $7:9 = 7x:9x$

Let

$$[7x * 4 + 9x * 3] * 2 = 220$$

$$28x + 27x = 110$$

$$55x = 110$$

$$x = 2$$

Let

The number of men in working in task 5 = $7x = 14$

11. Questions

Industry A

The total movies released in 2021 = 120

The number of hit movies released in 2021 = 25% of 120 = 30

The number of flop movies released in 2021 = $120 - 30 = 90$

The total movies released in 2022 = 75% of 120 = 90

Industry B

The total movies released in 2021 = 100

@QuizByMickey

The number of hit movies released in 2021 = 30% of 100 = 30

The number of flop movies released in 2021 = $100 - 30 = 70$

The total movies released in 2022 = 128% of 100 = 128

Industry C

The total movies released in 2021 = 80

The number of hit movies released in 2021 = 35% of 80 = 28

The number of flop movies released in 2021 = $80 - 28 = 52$

The total movies released in 2022 = 85% of 80 = 68

Industry D

The total movies released in 2021 = 160

The number of hit movies released in 2021 = 45% of 160 = 72

The number of flop movies released in 2021 = $160 - 72 = 88$

The total movies released in 2022 = 122.5% of 160 = 196

Industry E

The total movies released in 2021 = 60

The number of hit movies released in 2021 = 20% of 60 = 12

The number of flop movies released in 2021 = $60 - 12 = 48$

The total movies released in 2022 = 105% of 60 = 63

Let table

Industry	The total movies released in 2021	The number of hit movies released in 2021	The number of flop movies released in 2021	The total movies released in 2022
A	120	30	90	90
B	100	30	70	128
C	80	28	52	68
D	160	72	88	196
E	60	12	48	63

Answer: A

According to the question

The number of comedy movies released = 20% of 120 = 24

The number of horror movies released = 33.33% of $(120 - 24) = (1/3) * 96 = 32$

The number of action movies released = 75% of $(96 - 32) = 0.75 * 64 = 48$

The number of adventure movies released = $64 - 48 = 16$

Thus, difference = $90 - 16 = 74$

12. Questions

Industry A

The total movies released in 2021 = 120

The number of hit movies released in 2021 = 25% of 120 = 30

The number of flop movies released in 2021 = $120 - 30 = 90$

The total movies released in 2022 = 75% of 120 = 90

Industry B

The total movies released in 2021 = 100

The number of hit movies released in 2021 = 30% of 100 = 30

The number of flop movies released in 2021 = $100 - 30 = 70$

The total movies released in 2022 = 128% of 100 = 128

Industry C

The total movies released in 2021 = 80

The number of hit movies released in 2021 = 35% of 80 = 28

The number of flop movies released in 2021 = $80 - 28 = 52$

The total movies released in 2022 = 85% of 80 = 68

Industry D

The total movies released in 2021 = 160

The number of hit movies released in 2021 = 45% of 160 = 72

The number of flop movies released in 2021 = $160 - 72 = 88$

The total movies released in 2022 = 122.5% of 160 = 196

Industry E

The total movies released in 2021 = 60

The number of hit movies released in 2021 = 20% of 60 = 12

The number of flop movies released in 2021 = $60 - 12 = 48$

The total movies released in 2022 = 105% of 60 = 63

Let table

@QuizByMickey

Industry	The total movies released in 2021	The number of hit movies released in 2021	The number of flop movies released in 2021	The total movies released in 2022
A	120	30	90	90
B	100	30	70	128
C	80	28	52	68
D	160	72	88	196
E	60	12	48	63

Answer: C

The total number of flop movies released in 2022 in industries B and E together

$$= 75\% \text{ of } 128 + (100 - 22.22)\% \text{ of } 63$$

$$= 96 + (7/9) * 63$$

$$= 96 + 49$$

$$= 145$$

The total movies released in industries A and C together in 2021 = $120 + 80 = 200$

Thus, percentage = $(145/200) * 100 = 72.5\%$

13. Questions

Industry A

The total movies released in 2021 = 120

The number of hit movies released in 2021 = $25\% \text{ of } 120 = 30$

The number of flop movies released in 2021 = $120 - 30 = 90$

The total movies released in 2022 = $75\% \text{ of } 120 = 90$

Industry B

The total movies released in 2021 = 100

The number of hit movies released in 2021 = $30\% \text{ of } 100 = 30$ @QuizByMickey

The number of flop movies released in 2021 = $100 - 30 = 70$

The total movies released in 2022 = $128\% \text{ of } 100 = 128$

Industry C

The total movies released in 2021 = 80

The number of hit movies released in 2021 = $35\% \text{ of } 80 = 28$

The number of flop movies released in 2021 = $80 - 28 = 52$

The total movies released in 2022 = $85\% \text{ of } 80 = 68$

Industry D

The total movies released in 2021 = 160

The number of hit movies released in 2021 = 45% of 160 = 72

The number of flop movies released in 2021 = 160 – 72 = 88

The total movies released in 2022 = 122.5% of 160 = 196

Industry E

The total movies released in 2021 = 60

The number of hit movies released in 2021 = 20% of 60 = 12

The number of flop movies released in 2021 = 60 – 12 = 48

The total movies released in 2022 = 105% of 60 = 63

Let table

Industry	The total movies released in 2021	The number of hit movies released in 2021	The number of flop movies released in 2021	The total movies released in 2022
A	120	30	90	90
B	100	30	70	128
C	80	28	52	68
D	160	72	88	196
E	60	12	48	63

Answer: D

The total number of hit movies released in 2021 in all industries together = 30 + 30 + 28 + 72 + 12 = 172

The total number of flop movies released in 2021 in all industries together = 90 + 70 + 52 + 88 + 48 = 348

Thus, ratio = 172: 348 = 86:174 = 43: 87

14. Questions

Industry A

The total movies released in 2021 = 120

The number of hit movies released in 2021 = 25% of 120 = 30

The number of flop movies released in 2021 = 120 – 30 = 90

The total movies released in 2022 = 75% of 120 = 90

Industry B

The total movies released in 2021 = 100

The number of hit movies released in 2021 = 30% of 100 = 30

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The number of flop movies released in 2021 = $100 - 30 = 70$

The total movies released in 2022 = 128% of 100 = 128

Industry C

The total movies released in 2021 = 80

The number of hit movies released in 2021 = 35% of 80 = 28

The number of flop movies released in 2021 = $80 - 28 = 52$

The total movies released in 2022 = 85% of 80 = 68

Industry D

The total movies released in 2021 = 160

The number of hit movies released in 2021 = 45% of 160 = 72

The number of flop movies released in 2021 = $160 - 72 = 88$

The total movies released in 2022 = 122.5% of 160 = 196

Industry E

The total movies released in 2021 = 60

The number of hit movies released in 2021 = 20% of 60 = 12

The number of flop movies released in 2021 = $60 - 12 = 48$

The total movies released in 2022 = 105% of 60 = 63

Let table

Industry	The total movies released in 2021	The number of hit movies released in 2021	The number of flop movies released in 2021	The total movies released in 2022
A	120	30	90	90
B	100	30	70	128
C	80	28	52	68
D	160	72	88	196
E	60	12	48	63

Answer: B

The number of hit movies released in industry C in 2022 = $(68/17) * 10 = 40$

The number of flop movies released in industry C in 2022 = $(68/17) * 7 = 28$

The number of flop movies released in industry D in 2022 = $(28/40) * 100 = 70$

The number of hit movies released in industry D in 2022 = $196 - 70 = 126$

Thus, average = $(126 + 72)/2 = 198/2 = 99$

15. Questions

Industry A

The total movies released in 2021 = 120

The number of hit movies released in 2021 = 25% of 120 = 30

The number of flop movies released in 2021 = $120 - 30 = 90$

The total movies released in 2022 = 75% of 120 = 90

Industry B

The total movies released in 2021 = 100

The number of hit movies released in 2021 = 30% of 100 = 30

The number of flop movies released in 2021 = $100 - 30 = 70$

The total movies released in 2022 = 128% of 100 = 128

Industry C

The total movies released in 2021 = 80

The number of hit movies released in 2021 = 35% of 80 = 28

The number of flop movies released in 2021 = $80 - 28 = 52$

The total movies released in 2022 = 85% of 80 = 68

Industry D

The total movies released in 2021 = 160

The number of hit movies released in 2021 = 45% of 160 = 72

The number of flop movies released in 2021 = $160 - 72 = 88$

The total movies released in 2022 = 122.5% of 160 = 196

Industry E

The total movies released in 2021 = 60

The number of hit movies released in 2021 = 20% of 60 = 12

The number of flop movies released in 2021 = $60 - 12 = 48$

The total movies released in 2022 = 105% of 60 = 63

Let table

@QuizByMickey

Industry	The total movies released in 2021	The number of hit movies released in 2021	The number of flop movies released in 2021	The total movies released in 2022
A	120	30	90	90
B	100	30	70	128
C	80	28	52	68
D	160	72	88	196
E	60	12	48	63

Answer: E

The number of hit movies released in industry F in 2021 = 175% of 12 = 21

The total movies released in industry F in 2021 = $(80 + 160 + 60)/3 = 300/3 = 100$

The number of flop movies released in industry F in 2021 = $100 - 21 = 79$

16. Questions

Item A

The cost price = $24 * 100 = \text{Rs. } 2400$

We don't know the marked up percentage, so we cannot find the marked price and selling price

Item B

The cost price = $28 * 100 = \text{Rs. } 2800$

The profit amount of item B is Rs. 560.

The selling price = $2800 + 560 = \text{Rs. } 3360$

The profit percentage = $[560/2800] * 100 = 20\%$

$Y = 20$

Let

The marked price = $(3360/75) * 100 = \text{Rs. } 4480$

The discount amount = $4480 - 3360 = \text{Rs. } 1120$

Item C

The cost price = $16 * 100 = \text{Rs. } 1600$

We don't know the marked up percentage, so we cannot find the marked price and selling price

Item D

The cost price = $30 * 100 = \text{Rs. } 3000$

The profit percentage = $(Y + 33)\% = 53\%$

The selling price = $153\% \text{ of } 3000 = \text{Rs. } 4590$

@QuizByMickey

Let

The marked price = $(4590/85) * 100 = \text{Rs. } 5400$

The profit amount = $4590 - 3000 = \text{Rs. } 1590$

The discount amount = $5400 - 4590 = \text{Rs. } 810$

Item E

The cost price = $20 * 100 = \text{Rs. } 2000$

The profit percentage = $(Y + 12)\% = 32\%$

The selling price = 132% of $2000 = \text{Rs. } 2640$

Let

The marked price = $(2640/80) * 100 = \text{Rs. } 3300$

The discount amount = $3300 - 2640 = \text{Rs. } 660$

Let

Item	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)	The profit/loss (Rs.)	The discount (Rs.)
A	2400	-	-	-	-
B	2800	4480	3360	560	1120
C	1600	-	-	-	-
D	3000	5400	4590	1590	810
E	2000	3300	2640	640	660

Answer: B

According to the question

The value of $X = (20/2) * 1 = 10$

The discount percentage = $-X\% = -10\%$

The selling price of item A = $2400 * (90/100) = \text{Rs. } 2160$

Thus, difference = $3360 - 2160 = \text{Rs. } 1200$

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17. Questions

Item A

The cost price = $24 * 100 = \text{Rs. } 2400$

We don't know the marked up percentage, so we cannot find the marked price and selling price

Item B

The cost price = $28 * 100 = \text{Rs. } 2800$

The profit amount of item B is Rs. 560.

The selling price = $2800 + 560 = \text{Rs. } 3360$

The profit percentage = $[560/2800] * 100 = 20\%$

$Y = 20$

Let

The marked price = $(3360/75) * 100 = \text{Rs. } 4480$

The discount amount = $4480 - 3360 = \text{Rs. } 1120$

Item C

The cost price = $16 * 100 = \text{Rs. } 1600$

We don't know the marked up percentage, so we cannot find the marked price and selling price

Item D

The cost price = $30 * 100 = \text{Rs. } 3000$

The profit percentage = $(Y + 33)\% = 53\%$

The selling price = 153% of $3000 = \text{Rs. } 4590$

Let

The marked price = $(4590/85) * 100 = \text{Rs. } 5400$

The profit amount = $4590 - 3000 = \text{Rs. } 1590$

The discount amount = $5400 - 4590 = \text{Rs. } 810$

Item E

The cost price = $20 * 100 = \text{Rs. } 2000$

The profit percentage = $(Y + 12)\% = 32\%$

The selling price = 132% of $2000 = \text{Rs. } 2640$

Let

The marked price = $(2640/80) * 100 = \text{Rs. } 3300$

The discount amount = $3300 - 2640 = \text{Rs. } 660$

Let

@QuizByMickey

Item	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)	The profit/loss (Rs.)	The discount (Rs.)
A	2400	-	-	-	-
B	2800	4480	3360	560	1120
C	1600	-	-	-	-
D	3000	5400	4590	1590	810
E	2000	3300	2640	640	660

Answer: E

The marked-up percentage of item D = $[(5400 - 3000)/3000] * 100 = (2400/30) = 80\%$

The marked-up percentage of item E = $[(3300 - 2000)/2000] * 100 = (1300/20) = 65\%$

Thus, ratio = 80: 65 = 16: 13

18. Questions

Item A

The cost price = $24 * 100 = \text{Rs. } 2400$

We don't know the marked up percentage, so we cannot find the marked price and selling price

Item B

The cost price = $28 * 100 = \text{Rs. } 2800$

The profit amount of item B is Rs. 560.

The selling price = $2800 + 560 = \text{Rs. } 3360$

The profit percentage = $[560/2800] * 100 = 20\%$

$Y = 20$

Let

The marked price = $(3360/75) * 100 = \text{Rs. } 4480$

The discount amount = $4480 - 3360 = \text{Rs. } 1120$

Item C

The cost price = $16 * 100 = \text{Rs. } 1600$

We don't know the marked up percentage, so we cannot find the marked price and selling price

Item D

The cost price = $30 * 100 = \text{Rs. } 3000$

The profit percentage = $(Y + 33)\% = 53\%$

The selling price = 153% of $3000 = \text{Rs. } 4590$

@QuizByMickey

Let

The marked price = $(4590/85) * 100 = \text{Rs. } 5400$

The profit amount = $4590 - 3000 = \text{Rs. } 1590$

The discount amount = $5400 - 4590 = \text{Rs. } 810$

Item E

The cost price = $20 * 100 = \text{Rs. } 2000$

The profit percentage = $(Y + 12)\% = 32\%$

The selling price = 132% of $2000 = \text{Rs. } 2640$

Let

The marked price = $(2640/80) * 100 = \text{Rs. } 3300$

The discount amount = $3300 - 2640 = \text{Rs. } 660$

Let

Item	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)	The profit/loss (Rs.)	The discount (Rs.)
A	2400	-	-	-	-
B	2800	4480	3360	560	1120
C	1600	-	-	-	-
D	3000	5400	4590	1590	810
E	2000	3300	2640	640	660

Answer: A

The discount amount of item C = 27.5% of $640 = \text{Rs. } 176$

The discount percentage of item C is 10%

The selling price of item C = $(176/10) * 90 = \text{Rs. } 1584$

The profit/loss percentage of item C = $[(1600 - 1584)/1600] * 100 = (16/16) = 1\%$

So, $X/10 = 1\%$

$X = 10\%$

The loss percentage of item A is 10%

The selling price of item A = 90% of $2400 = \text{Rs. } 2160$

The marked price of item A = $(2160/72) * 100 = \text{Rs. } 3000$

The marked up percentage of item A = $[(3000 - 2400)/2400] * 100 = 25\%$

19. Questions

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Item A

The cost price = $24 * 100 = \text{Rs. } 2400$

We don't know the marked up percentage, so we cannot find the marked price and selling price

Item B

The cost price = $28 * 100 = \text{Rs. } 2800$

The profit amount of item B is Rs. 560.

The selling price = $2800 + 560 = \text{Rs. } 3360$

The profit percentage = $[560/2800] * 100 = 20\%$

$Y = 20$

Let

The marked price = $(3360/75) * 100 = \text{Rs. } 4480$

The discount amount = $4480 - 3360 = \text{Rs. } 1120$

Item C

The cost price = $16 * 100 = \text{Rs. } 1600$

We don't know the marked up percentage, so we cannot find the marked price and selling price

Item D

The cost price = $30 * 100 = \text{Rs. } 3000$

The profit percentage = $(Y + 33)\% = 53\%$

The selling price = $153\% \text{ of } 3000 = \text{Rs. } 4590$

Let

The marked price = $(4590/85) * 100 = \text{Rs. } 5400$

The profit amount = $4590 - 3000 = \text{Rs. } 1590$

The discount amount = $5400 - 4590 = \text{Rs. } 810$

Item E

The cost price = $20 * 100 = \text{Rs. } 2000$

The profit percentage = $(Y + 12)\% = 32\%$

The selling price = $132\% \text{ of } 2000 = \text{Rs. } 2640$

Let

The marked price = $(2640/80) * 100 = \text{Rs. } 3300$

The discount amount = $3300 - 2640 = \text{Rs. } 660$

Let

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Item	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)	The profit/loss (Rs.)	The discount (Rs.)
A	2400	-	-	-	-
B	2800	4480	3360	560	1120
C	1600	-	-	-	-
D	3000	5400	4590	1590	810
E	2000	3300	2640	640	660

Answer: C

The cost price of item F = 120% of 1600 = Rs. 1920

The marked up percentage of item D = $[(5400 - 3000)/3000] * 100 = 80\%$

The marked price of item F = 180% of 1920 = Rs. 3456

The selling price of item F = 105% of 1920 = Rs. 2016

The discount amount of item F = 3456 – 2016 = Rs. 1440

Thus, sum = 1440 + 810 = Rs. 2250

20. Questions

Item A

The cost price = $24 * 100 = \text{Rs. } 2400$

We don't know the marked up percentage, so we cannot find the marked price and selling price

Item B

The cost price = $28 * 100 = \text{Rs. } 2800$

The profit amount of item B is Rs. 560.

The selling price = $2800 + 560 = \text{Rs. } 3360$

The profit percentage = $[560/2800] * 100 = 20\%$

Y = 20

Let

The marked price = $(3360/75) * 100 = \text{Rs. } 4480$

The discount amount = $4480 - 3360 = \text{Rs. } 1120$

Item C

The cost price = $16 * 100 = \text{Rs. } 1600$

We don't know the marked up percentage, so we cannot find the marked price and selling price

Item D

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The cost price = $30 \times 100 = \text{Rs. } 3000$

The profit percentage = $(Y + 33)\% = 53\%$

The selling price = 153% of $3000 = \text{Rs. } 4590$

Let

The marked price = $(4590/85) \times 100 = \text{Rs. } 5400$

The profit amount = $4590 - 3000 = \text{Rs. } 1590$

The discount amount = $5400 - 4590 = \text{Rs. } 810$

Item E

The cost price = $20 \times 100 = \text{Rs. } 2000$

The profit percentage = $(Y + 12)\% = 32\%$

The selling price = 132% of $2000 = \text{Rs. } 2640$

Let

The marked price = $(2640/80) \times 100 = \text{Rs. } 3300$

The discount amount = $3300 - 2640 = \text{Rs. } 660$

Let

Item	The cost price (Rs.)	The marked price (Rs.)	The selling price (Rs.)	The profit/loss (Rs.)	The discount (Rs.)
A	2400	-	-	-	-
B	2800	4480	3360	560	1120
C	1600	-	-	-	-
D	3000	5400	4590	1590	810
E	2000	3300	2640	640	660

Answer: A

(The sum of marked price of items A and C together + $4480 + 5400 + 3300$) - $(2400 + 2800 + 1600 + 3000 + 2000) = 6140$

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The sum of marked price of items A and C together + $1380 = 6140$

The sum of marked price of items A and C together = 4760

The marked price of item A is Rs. 1240 more than that of item C

The marked price of item A = $(4760 + 1240)/2 = \text{Rs. } 3000$

The marked price of item C = $4760 - 3000 = \text{Rs. } 1760$

The selling price of item A = 72% of $3000 = \text{Rs. } 2160$

The loss amount of item A = $2400 - 2160 = \text{Rs. } 240$

The selling price of item C = 90% of 1760 = Rs. 1584

The loss amount of item C = 1600 – 1584 = Rs. 16

Thus, the loss amount of items A and C together = 240 + 16 = Rs. 256

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